Exp 8

Program:

```
import cv2
import matplotlib.pyplot as plt
from IPython.display import display, clear_output

cap = cv2.VideoCapture("images/motiontracking/car.mp4")

ret, frame1 = cap.read()
if not ret:
print("Error: Cannot read video")
exit()

prev_gray = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)
prev_gray = cv2.GaussianBlur(prev_gray, (11, 11), 0)

print("Car Motion Tracking Started...")

while True:
ret, frame2 = cap.read()
if not ret:
break

gray = cv2.cvtColor(frame2, cv2.COLOR_BGR2GRAY)
gray = cv2.GaussianBlur(gray, (11, 11), 0)
```

```
diff = cv2.absdiff(prev_gray, gray)

_, thresh = cv2.threshold(diff, 20, 255, cv2.THRESH_BINARY)

thresh = cv2.dilate(thresh, None, iterations=2)
contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

for c in contours:
if cv2.contourArea(c) > 1000:
x, y, w, h = cv2.boundingRect(c)
cv2.rectangle(frame2, (x, y), (x + w, y + h), (0, 255, 0), 2)

frame_rgb = cv2.cvtColor(frame2, cv2.COLOR_BGR2RGB)
clear_output(wait=True)
plt.imshow(frame_rgb)
plt.title("Car Motion Tracking")
plt.axis('off')
display(plt.gcf())
plt.close()
prev_gray = gray
cap.release()
print("Tracking Stopped")
```

```
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.title("Car Motion Tracking")
    plt.axis('off')
    display(plt.gcf())
    plt.close()

    # Update previous frame
    prev_gray = gray

# Release video
cap.release()
print("Tracking Stopped")
```